


# ISLAMIC UNIVERSITY OF TECHNOLOGY

## Department of Electrical and Electronic Engineering

## EEE 4418 Project Report

## ELECTRICAL LAYOUT DESIGN OF A FOUR-STORIED RESIDENTIAL BUILDING

**Submitted by**

**Group-4(C1)**


| | |
|---|---|
| Ahnaf Kaiser | 230021301 |
| Ishrak Samin | 230021303 |
| Nasif Faiyaz | 230021305 |
| Shaheen Rahman Dibya | 230021315 |

# TABLE OF CONTENTS


**1. Introduction**

**2. Floor Plans**

- 2.1 Typical Floor Plan
- 2.2 Unit Layout — Unit A
- 2.3 Ground Floor — Parking and Services
- 2.4 Roof Top Plan

**3. Electrical Fittings and Fixtures**

- 3.1 Typical Floor — Fixture Layout
- 3.2 Unit-Wise Fixture Schedule

**4. Load Calculation**

- 4.1 Apparatus Schedule and Assumed Ratings
- 4.2 Branch Distribution Board (BDB) — Circuit Currents for One Unit
- 4.3 Total BDB Load and Incomer Cable Selection
- 4.4 Emergency Branch Distribution Board (EBDB) — Load and Current for One Unit
- 4.5 Main Distribution Board (MDB) — Building Load and Breaker Sizing

**5. Distribution Boards and Single-Line Diagrams**

- 5.1 Switch Boards (SB)
- 5.2 Branch Distribution Boards (BDB)
- 5.3 Emergency Switch Boards (ESB)
- 5.4 Emergency Branch Distribution Board (EBDB)
- 5.5 Main Distribution Board (MDB)
- 5.6 Emergency Main Distribution Board (EMDB)

**6. Conduit and Cable Selection**

- 6.1 Point-Wiring Conduits
- 6.2 Circuit and Feeder Conductors

**7. Auxiliary Systems**

- 7.1 Telephone, Television and Data Network
- 7.2 Lightning Protection System

**8. Summary Table**

**9. Conclusion**

# 1. Introduction

This report presents the complete electrical layout design of a four-storied residential building, developed as part of our EEE 441 course. The entire design was prepared using AutoCAD and includes the architectural planning of the building, placement of electrical fittings and fixtures, wiring and conduit routing, connected-load calculations, sizing of distribution boards, and selection of suitable cables and conduits. In addition to the main electrical system, two auxiliary systems were also designed: a telephone/TV/data network and a lightning protection system.

The building has a ground floor mainly used for vehicle parking and utility services, including a generator room, gas cylinder room, and guard room. Above the ground floor, there are four typical residential floors, along with an open rooftop containing the water tank and stair head. Each residential floor is divided into two mirror-image units, giving a total of eight residential units in the building. The electrical power is supplied through a three-phase Main Distribution Board (MDB), which feeds one Branch Distribution Board (BDB) for each residential unit, along with separate boards for the parking floor and rooftop. A separate generator-backed emergency network is also provided, consisting of an Emergency Main Distribution Board (EMDB), Emergency Branch Distribution Boards (EBDB), and Emergency Switch Boards (ESB). These emergency boards are designed to supply essential lighting and exhaust points throughout the building whenever there is a failure in the main power supply.

Since all eight residential units have the same layout, mirrored around the central lobby and staircase on each floor, only one representative unit, referred to as Unit A throughout this report, is described in full detail. The remaining units follow the same architectural and electrical arrangement. The main project parameters and figures are summarized in Table 1 below.

**Table 1: Project at a glance**

| Parameter | Value |
|---|---|
| Number of residential (typical) floors | 4 |
| Additional levels | 1 ground floor (parking) + 1 rooftop |
| Residential units per floor | 2 |
| Total residential units | 8 |
| Total branch/distribution boards (normal + emergency) | 21 |
| Total connected electrical load | 61.6 kW |

The remainder of this report is organized into several sections, each covering a specific part of the overall design. Section 2 describes the architectural floor plans, while Section 3 presents the layout of the electrical fittings and fixtures. Section 4 focuses on the connected-load calculations, followed by the distribution-board single-line diagrams in Section 5. Section 6 discusses the selection of cables and conduits. The two auxiliary systems are described in Section 7, and Section 8 provides a building-wide summary of the major project details. Finally, Section 9 concludes the report.

# 2. Floor Plans

The architectural layout was drawn in AutoCAD before any electrical elements were added, so that room sizes and circulation paths could be fixed first. This section presents the typical floor plan, the layout of one representative residential unit, the ground floor (parking and services), and the rooftop plan.

## 2.1 Typical Floor Plan

Each typical floor follows a symmetrical layout, with two identical residential units, Unit A and Unit B, arranged as mirror images of each other. Both units open into a shared central lobby, with the staircase located in the middle along the building's axis of symmetry. Since the two units have the same arrangement, only Unit A is described room-by-room in Section 2.2. Unit B simply follows the same layout in mirror-image form across the central corridor. Figure 1 shows the typical floor plan prepared in AutoCAD, while Table 2 provides its main dimensions

**Table 2: Typical floor — key dimensions**

| Feature | Dimension |
|---|---|
| Overall floor footprint | 60′-0″ × 35′-7.5″ |
| Central lobby | 33′-11.5″ × 8′-1.5″ |
| Residential units per floor | 2 |

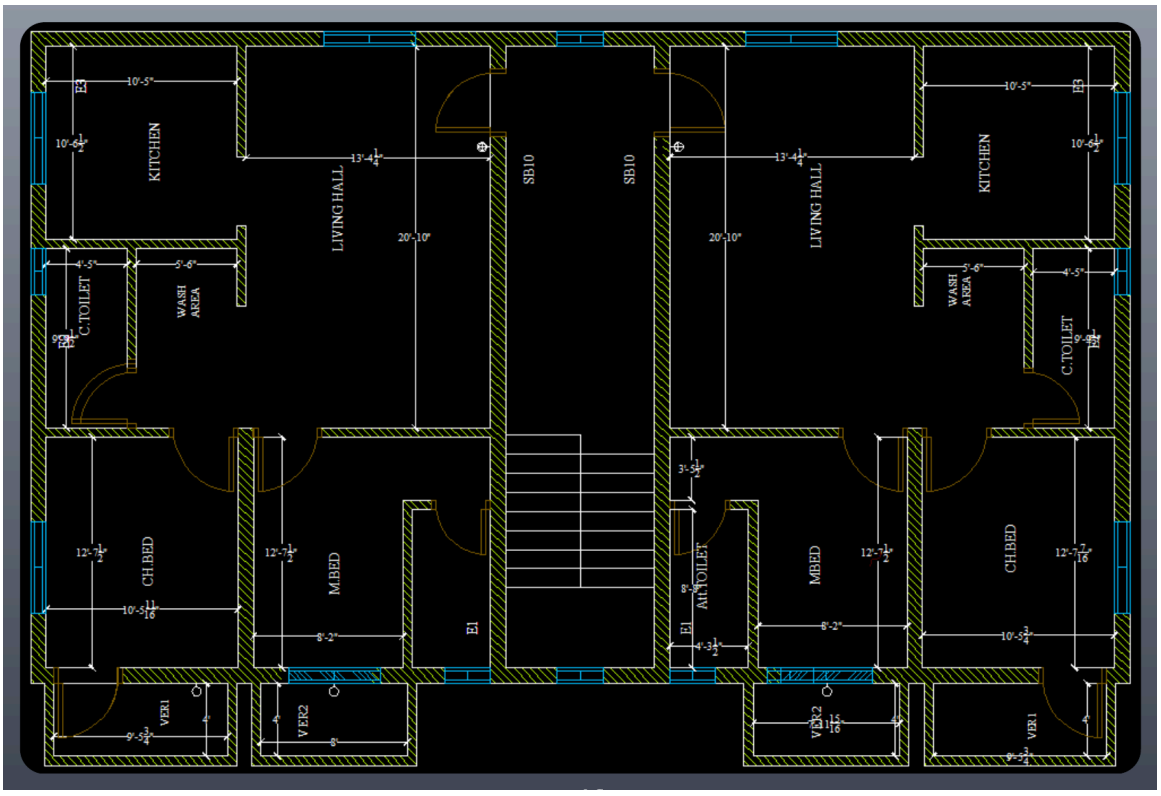



Figure 1: Typical floor plan, showing the two mirrored residential units, the central lobby and the staircase core.

## 2.2 Unit Layout — Unit A

Unit A comprises a master bedroom with an attached toilet and a private verandah, a child's bedroom with access to a second verandah and served by a common toilet, a living hall that acts as the internal

circulation/distribution space of the unit, and a kitchen placed against the outer wall for cross-ventilation. The room schedule is given in Table 3, and the enlarged unit layout is shown in Figure 2.

**Table 3: Unit A — room schedule**

| Room | Dimensions |
|---|---|
| Master bedroom | 12′-7.5″ × 8′-2″ |
| Child's bedroom | 12′-7.5″ × 10′-5.7″ |
| Living hall | 20′-10″ × 13′-4.25″ |
| Kitchen | 10′-6.5″ × 10′-5″ |
| Attached toilet | 8′-8″ × 4′-3.5″ |
| Common toilet | 9′-9.5″ × 4′-5″ |

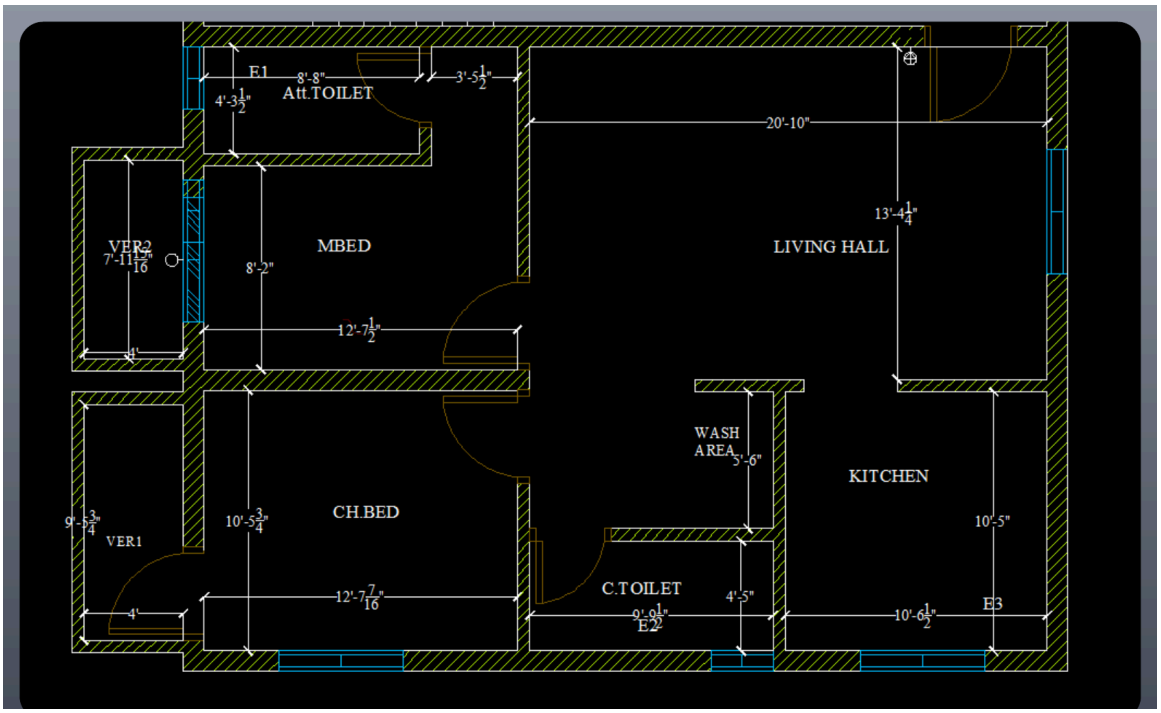



Figure 2: Unit A layout — representative of both units on every typical floor.

**Circulation notes**

- A private verandah is attached to both the master bedroom and the child's bedroom.
- The kitchen is placed on the outer wall of the unit to allow cross-ventilation and exhaust.
- The attached toilet serves the master bedroom; the common toilet serves the child's bedroom.
- The living hall acts as the internal distribution space of the unit, connecting all rooms to the main entrance.

## 2.3 Ground Floor — Parking and Services

The ground floor is mainly used for vehicle parking and building services rather than residential purposes. It includes four car-parking spaces, a generator room, a gas-cylinder room, and a guard room with an attached toilet. There are two entrances from the street, while the staircase core is located on

the west side of the building and continues to the floors above. A vehicle ramp on the same side provides access to the parking level. The main dimensions of the rooms are given in Table 4, and the complete floor layout is shown in Figure 3.

**Table 4: Ground floor — room and provision dimensions**

| Provision | Dimensions |
|---|---|
| Car park (each of 4 bays) | 16′-0″ × 9′-5.4″ |
| Generator room | 10′-9.5″ × 9′-4.5″ |
| Gas cylinder room | 10′-9.5″ × 4′-8.9″ |
| Guard room | 10′-11.1″ × 5′-6″ |
| Guard toilet | 7′-11″ × 3′-11″ |

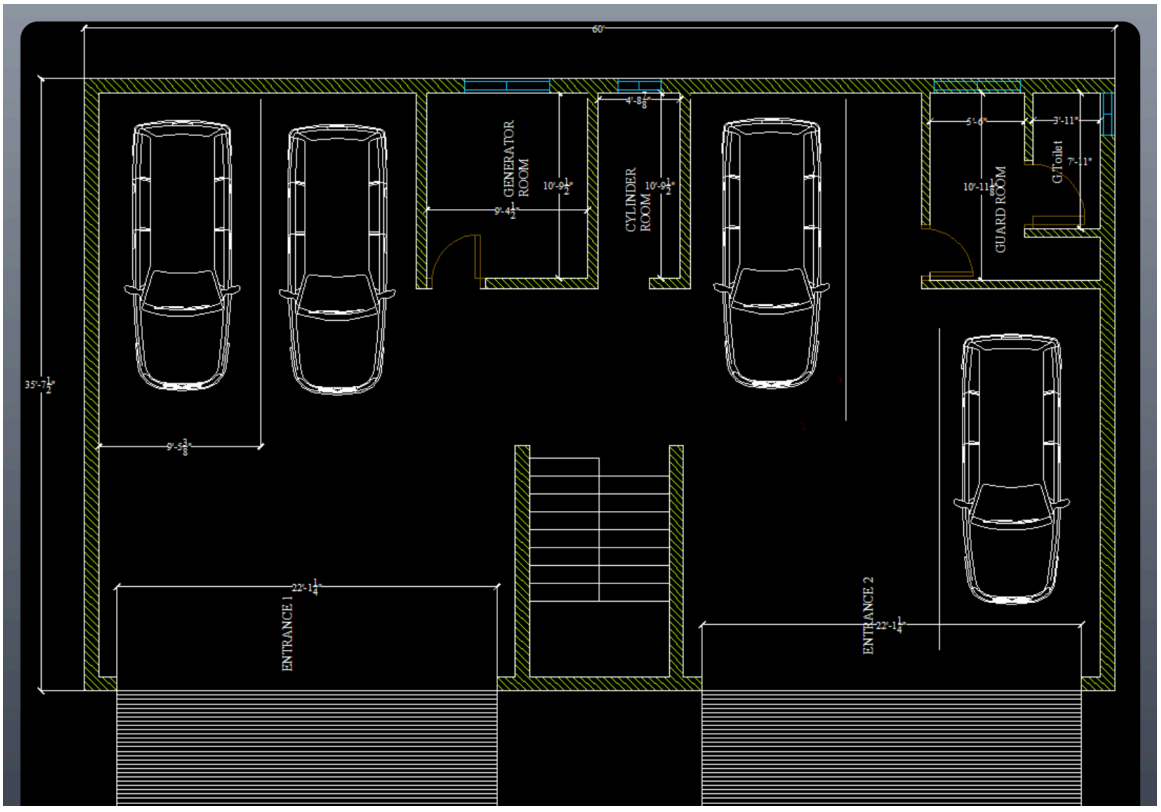



Figure 3: Ground floor plan — vehicle parking, generator room, gas-cylinder room and guard facilities.

**Design notes**

-The staircase is situated on the west side of the building, shared with all floors above.

- A ramp on the west side gives vehicle access down to the parking level.

-The generator room and gas-cylinder room are positioned well away from the guard room, for safety and to avoid disturbing the guard's accommodation.

- The guard room has a side window and its toilet has an exhaust window, ensuring adequate ventilation.

### 2.4 Roof Top Plan

The rooftop is left largely open for common use, apart from the covered stair head and the overhead water tank. The staircase terminates at a covered stair head that opens onto the roof, and the water tank is placed alongside it. Table 5 summarizes the rooftop provisions and Figure 4 shows the plan.

**Table 5: Roof top — key areas**

| Feature | Dimensions |
|---|---|
| Roof top free area | 60′-0″ × 35′-7.5″, less the water tank and stair-head area |
| Water tank area | 9′-5″ × 8′-11.9″ |

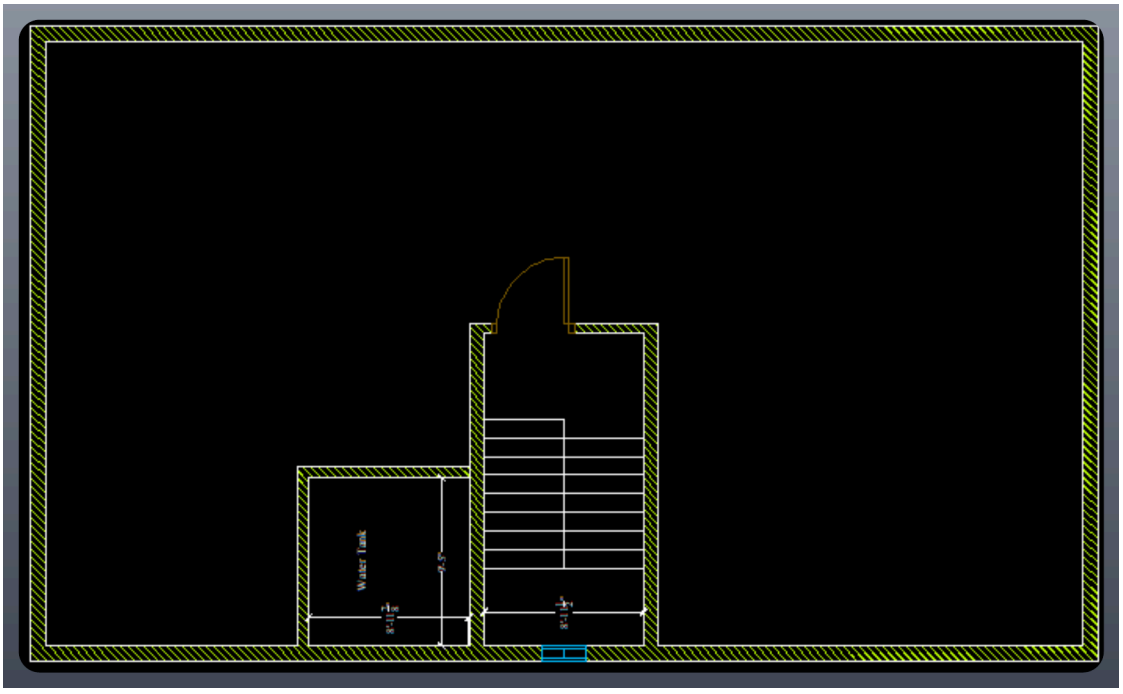



Figure 4: Roof top plan — water tank area and covered stair head.

**Design notes**

- The stair terminates in a covered stair head that gives access to the open roof.
- The remaining roof area is left open for common use by residents.

## 3. Electrical Fittings and Fixtures

With the architectural layout fixed, lighting points, fans, exhaust fans, switch boards and socket outlets were placed room-by-room in AutoCAD. A consistent symbol set (Table 6) was used throughout the drawings so that every fixture can be identified directly from the layout.

**Table 6: Electrical symbol legend**

| Symbol | Description |
|---|---|
| F | Ceiling fan |
| LB | Light bulb |
| CB | Ceiling bulb |
| SB | Switch board |
| SS | Switchboard socket — 2-pin socket outlet |
| E | Exhaust fan |
| ESB | Emergency board |
| P | 3-pin, 15 A socket outlet (3000 W) |

## 3.1 Typical Floor — Fixture Layout

Figure 5 shows the complete fixture layout for a typical floor, with every light point, fan, switch board, socket outlet and exhaust fan marked using the symbols in Table 6.

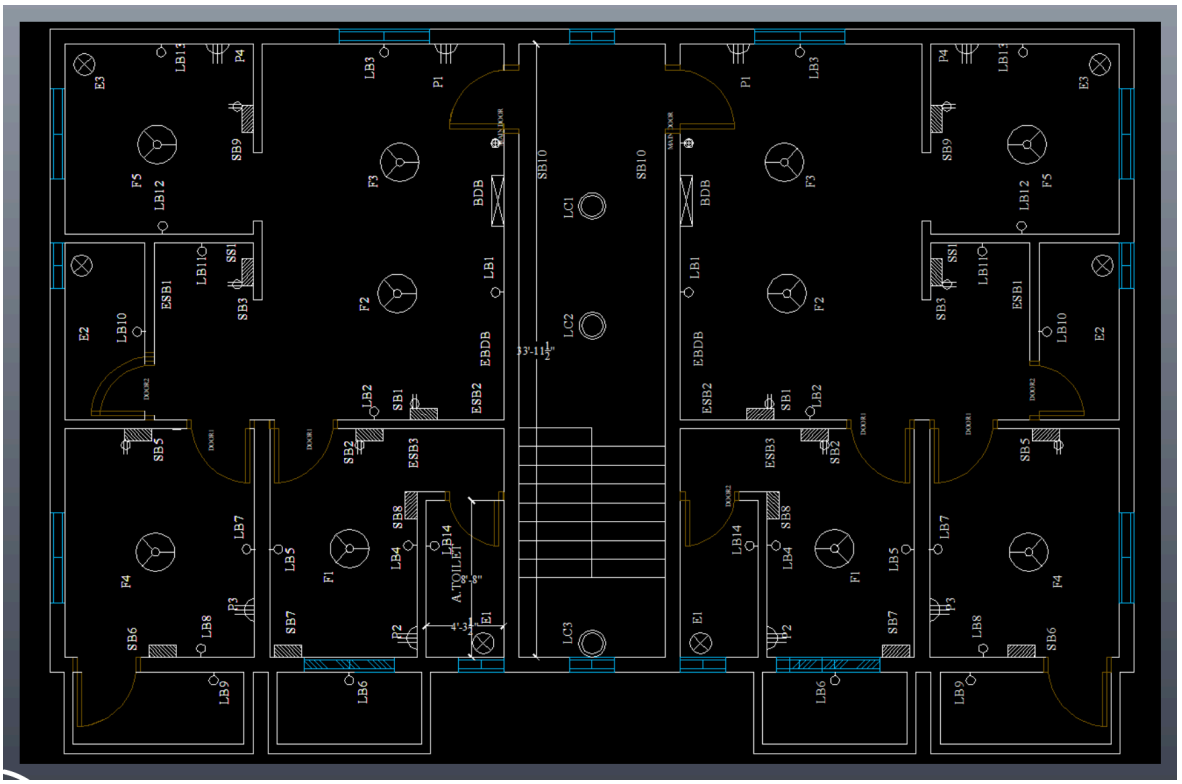


Figure 5: Typical floor — position of all lights, fans, switch boards and socket outlets.

## 3.2 Unit-Wise Fixture Schedule

Table 7 breaks down the fixture count of Unit A room-by-room; Figure 6 shows the corresponding enlarged layout. The living hall carries the largest fixture count, consistent with it also housing the unit's only television point and acting as the main internal distribution space.

**Table 7: Unit A — fixture schedule by room**

| Room | Light | Fan | Switch board | 3-pin socket | TV | Exhaust fan |
|---|---|---|---|---|---|---|
| Master bedroom | 2 | 1 | 1 | 1 | 0 | 0 |
| Child's bedroom | 2 | 1 | 1 | 1 | 0 | 0 |
| Living hall | 3 | 2 | 1 | 1 | 1 | 0 |
| Kitchen | 2 | 1 | 1 | 1 | 0 | 1 |
| Attached toilet | 1 | 0 | 1 | 0 | 0 | 1 |
| Common toilet | 1 | 0 | 1 | 0 | 0 | 1 |
| Verandahs (total) | 2 | 0 | 2 | 0 | 0 | 0 |
| **Total per unit** | **11** | **5** | **8** | **4** | **1** | **3** |

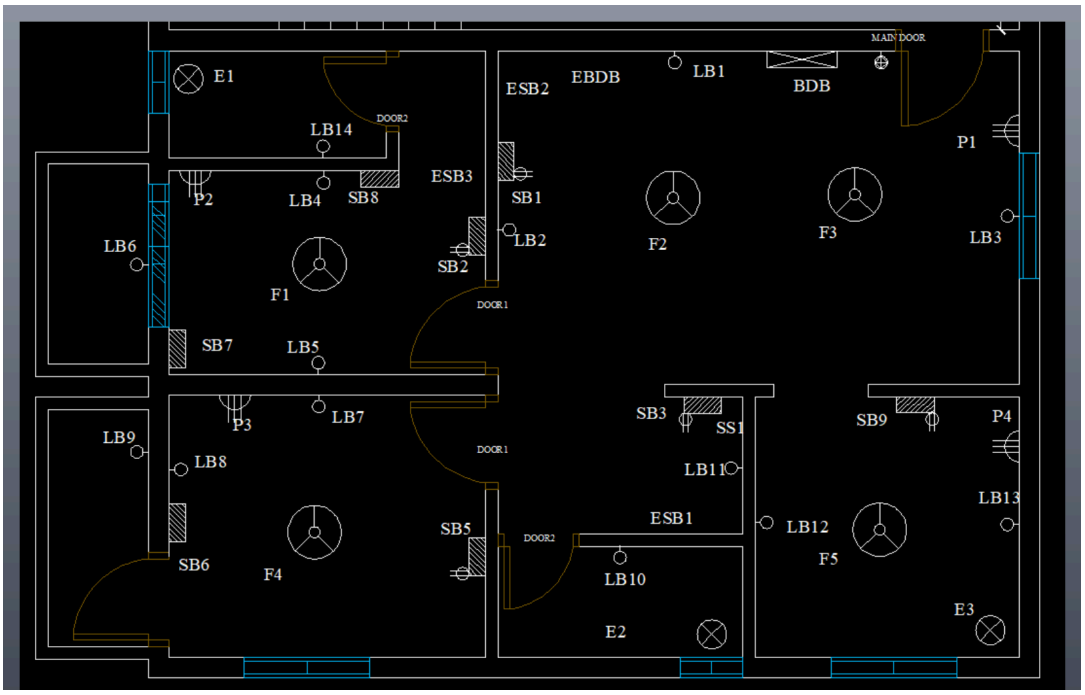



Figure 6: Unit A — enlarged view of the fitting and fixture layout.

**Board arrangement**

- Each room, verandah and toilet is given its own separate switch board, so that every room's fixtures can be controlled independently.
- Every switch board carries one 2-pin switchboard socket outlet (SS).
- In addition, every room is given its own 3-pin, 15 A socket outlet (P) for heavier appliances.

## 4. Load Calculation

The connected load was built up from the bottom of the distribution hierarchy upward: individual room fixtures were totaled into branch-circuit loads, branch circuits were combined (with demand factors) into each unit's Branch Distribution Board (BDB) load, and unit loads were combined with the parking floor, rooftop, pump and emergency system loads to size the building's Main Distribution Board (MDB).

Throughout this section, a power factor of cos φ = 0.7 is used to convert real power to current, consistent with the value adopted for the building's incoming supply.

## 4.1 Apparatus Schedule and Assumed Ratings

Table 8 lists the assumed wattage of each type of fixture, and Table 9 gives the resulting apparatus count and connected load per unit, per floor, and for the whole building.

**Table 8: Assumed fixture ratings**

| Fixture | Assumed rating |
|---|---|
| Ceiling fan (F) | 100 W |
| Light bulb (LB) | 20 W |
| Ceiling bulb (CB) | 20 W |
| Switchboard socket (SS) | 100 W |
| Exhaust fan (E) | 50 W |

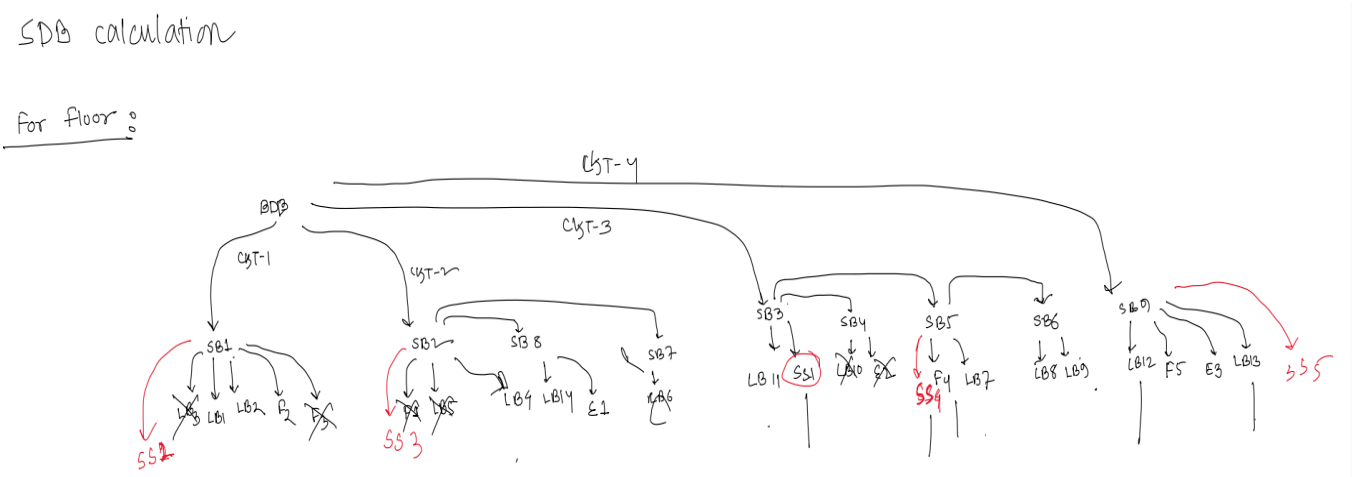
**Table 9: Apparatus schedule — per unit, per floor and building-wide**

| Apparatus | Rating | Per unit | Per floor | Building (8 units) |
|---|---|---|---|---|
| Light point (LB / CB / TB) | 20 W | 14 | 28 | 112 |
| Ceiling fan (F) | 100 W | 5 | 10 | 40 |
| Switch board (SB) | — | 8 | 16 | 80 |
| Switchboard socket (SS) | 100 W | 5 | 10 | 40 |
| 3-pin socket outlet (P) | 3000 W | 4 | 8 | 32 |
| Exhaust fan (E) | 50 W | 3 | 6 | 24 |

Across the whole building this gives 112 light points, 40 ceiling fans, 72 socket outlets (switchboard sockets and 3-pin outlets combined) and 24 exhaust fans. The number of light points in each room was fixed using the lumen method,

$$N = (E \times A) \div (n \times F \times UF \times LLF)$$

where N is the number of lighting points required, E is the required illuminance, A is the room area, n is the number of lamps per fitting, F is the luminous flux per lamp, UF is the utilisation factor and LLF is the light-loss factor.



power ratings

LB = 20W
LC = 20W
F = 100W
E = 50W
pf = 0.7 (average rating)
SS = 100 W (max)

CKT-1 calculations

$$I = \frac{2 \times 20 + 100 \times 1}{220 \times 0.7}$$

$$= 1.55A$$

∴ 2×1.5 sqmm BYM + 1.5 BYA ECC are used

CKT-2 calculation

$$I = \frac{3 \times 20 + 50 + 100}{220 \times 0.7}$$

$$= 1.36A$$

∴ 2×1.5 sqmm BYM + 1.5 BYA ECC are used

CKT-3 calculation

$$I = \frac{4 \times 20 + 100 + 100 + 100}{220 \times 0.7}$$

$$= 2.46A$$

∴ 2×1.5 sqmm BYM + 1.5 BYA ECC are used

CKT-4 calculation

$$I = \frac{2 \times 20 + 100 + 50 + 100}{220 \times 0.7}$$

$$= 1.86A$$

∴ 2×1.5 sqmm BYM + 1.5 BYA ECC are used

for Ground :




CKT-1 calc:

$$I = \frac{20}{220 \times 0.7}$$

$$= 0.12A$$

∴ 2×1.5 sqmm BYM + 1.5 BYA ECC are used

power ratings

LB = 20W
LC = 20W
F = 100W
E = 50W
pf = 0.7 (average rating)

CKT-2 calc:

$$I = \frac{7 \times 20}{220 \times 0.7}$$

$$= 0.9A$$

∴ 2×1.5 sqmm BYM + 1.5 BYA ECC are used

CKT-3 calc:

$$I = \frac{3 \times 20 + 50 + 100}{220 \times 0.7}$$

$$= 1.36A$$

∴ 2×1.5 sqmm BYM + 1.5 BYA ECC are used

Roof top:




CKT1 calc:

$$I = \frac{20 \times 2}{220 \times 0.7}$$

$$= 0.25A$$

2×1.5 sqmm BYM + 1.5 BYA ECC are used

## 4.2 Branch Distribution Board (BDB) — Circuit Currents for One Unit

Each unit's lighting and fan load is split across four circuits (CKT-1 to CKT-4) at its BDB, sized using:

$$I = P \div (V \cdot \cos\varphi), \quad V = 220\ V, \ \cos\varphi = 0.7$$

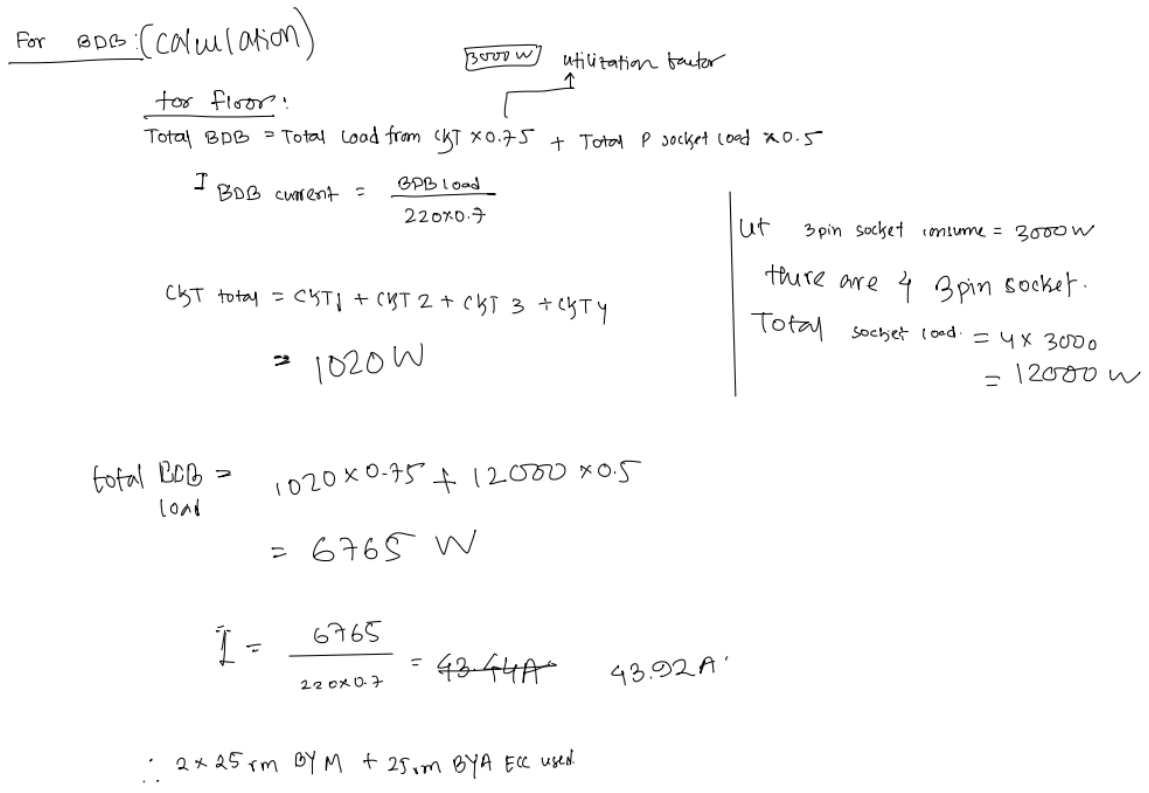
**Table 10: BDB circuit loads and currents — one unit**

| Circuit | Connected apparatus | Load (W) | Current (A) |
|---|---|---|---|
| CKT-1 | F + 2×LB + SS | 140 | 1.55 |
| CKT-2 | F + 3×LB + E + SS | 210 | 1.36 |
| CKT-3 | 4×LB + F + 2×SS | 380 | 2.46 |
| CKT-4 | F + E + 2×LB + SS | 290 | 1.88 |
| **Total connected circuit load** | | **1020** | **—** |

Worked example — CKT-3 (the heaviest of the four circuits):

$$I = (4 \times 20 + 100 + 2 \times 100) \div (220 \times 0.7) = 380 \div 154 \approx 2.46\ A$$

Since 2.46 A is the largest circuit current, a 5 A single-pole (SP) MCB is sufficient to protect every lighting/fan circuit in the unit. Each circuit is wired from its BDB in 2×1.5 mm² BYM + 1.5 mm² BYA (ECC) conductors.



for Ground: CKT total = ~~270~~ 370

$$I = \frac{370 \times 0.75}{220 \times 0.7}$$

$$= 1.80$$

∴ 2×1.5 sq.mm BYM + 1.5 sq.mm BYA used.

for rooftop:

CKT = 40 W

$$I = \frac{40 \times 0.75}{220 \times 0.7}$$

$$= 0.19 A$$

∴ 2×1.5 sq.mm BYM + 1.5 BYA ECC are used

### 4.3 Total BDB Load and Incomer Cable Selection

The four 3-pin socket outlets (P1–P4) in each unit are rated at 3000 W each and are wired as a separate group of socket circuits, protected together by a 50 A single-pole MCCB and wired in 2×6 mm² BYM + 6 mm² BYA (ECC). To estimate the maximum demand at the BDB incomer — rather than simply summing every connected watt, which would over-state the design current — a demand factor of 0.75 is applied to the connected lighting/fan circuit load, and a demand factor of 0.5 is applied to the socket load:

**Total BDB load = (Total circuit load × 0.75) + (Total socket load × 0.5)**

| Quantity | Value |
|---|---|
| Total connected circuit (lighting/fan) load | 1020 W |
| Total socket load (4 × 3000 W) | 12,000 W |
| Total BDB load = 1020×0.75 + 12,000×0.5 | 765 + 6000 = 6765 W |
| BDB incomer current, I = 6765 ÷ (220 × 0.7) | 43.92 A |

A current of 43.92 A calls for a cable of 2×25 mm² BYM + 25 mm² BYA (ECC), protected at the MDB end by a 50 A moulded-case circuit breaker (MCCB) — this feeder and breaker size is used for each of the eight identical unit BDBs.

### 4.4 Emergency Branch Distribution Board (EBDB) — Load and Current for One Unit

Each unit's emergency lighting network — one light point on the exhaust-fan circuit plus a light and fan on two further circuits — is fed from three emergency circuits (CKT-1′ to CKT-3′) at the unit's EBDB, sized with the same demand-factor approach:

**EBDB design load = (Total connected load × 0.75) + (Power-socket load × 0.5)**

**Table 11: EBDB circuit loads — one unit**

| Circuit | Connected apparatus | Load (W) |
|---|---|---|
| CKT-1′ | E + LB | 70 |
| CKT-2′ | F + LB | 120 |
| CKT-3′ | F + LB | 120 |
| **Total connected emergency load** | | **310** |

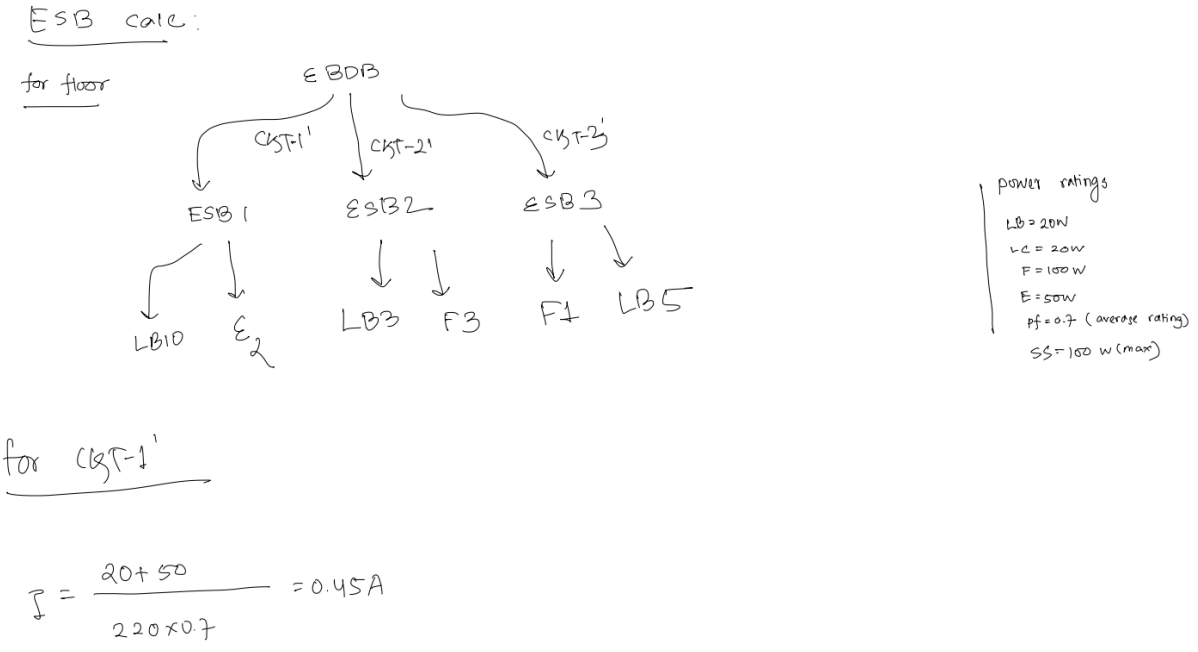
There is no socket load on the emergency network, so the design load is simply the connected load reduced by the 0.75 demand factor:

**Design load = 310 × 0.75 = 232.5 W  I = 232.5 ÷ (220 × 0.7) ≈ 1.50 A**

Each unit's EBDB is therefore wired back to the Emergency Main Distribution Board (EMDB) in 2×1.5 mm² BYM + 1.5 mm² BYA (ECC), with the three emergency circuits individually protected at 5 A.

**Building-wide emergency load**

| Segment | Quantity | Load |
|---|---|---|
| Residential floor (8 units × 232.5 W) | 8 | 1860 W |
| Lobby (per floor, ×4 floors) | 4 | 240 W |
| Basement / parking floor | 1 | 120 W |
| **Total EMDB design load** | | **2220 W** |



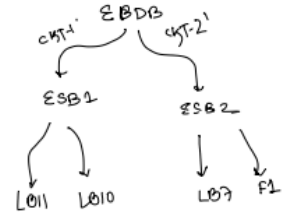
for CKT-2'

$$I = \frac{100+20}{220\times0.7} = 0.77A$$

For CKT-3'

$$I = \frac{100+20}{220\times0.7} = 0.77A$$

for ground:




CKT-1'

$$I = \frac{20\times2}{220\times0.7}$$

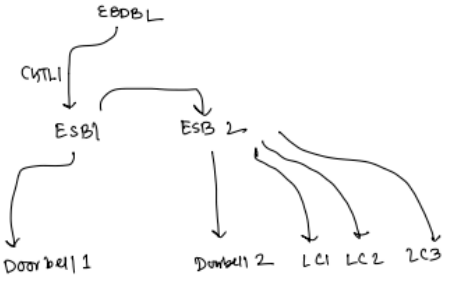
$$= 0.25A$$

CKT-2'

$$I = \frac{20+100}{220\times0.7}$$

$$= 0.77A$$

for lobby:




CKTL1

$$I = \frac{10\times2+20\times3}{220\times0.7} = 0.51A$$

For EBDB

for floor:

Total EBDB = Total Load × 0.75 → (CST-1' + EBT-2' + CBT-3')

= 310 × 0.75

= 232.5

$$I = \frac{232.5}{220 \times 0.7} = 1.50\,A$$

for Ground

Total EBDB = Total Load × 0.75

= 160 × 0.75 = 120W

$$I = \frac{120}{220 \times 0.7} = 0.77A$$

for Lobby

Total EBDB = 80 × 0.75

= 60W

$$I = \frac{60}{220 \times 0.7} = 0.38A$$

EMDB calculations

power = 232.5 × 8 + 60 × 4 + 120

(232.5 → per unit power; 8 → no of unit (4 floor); 60 → Lobby; 120 → Basement.)

= 2220 W

$P = 3VI\,pf \Rightarrow I = 4.80_A$

### 4.5 Main Distribution Board (MDB) — Building Load and Breaker Sizing

The MDB is a three-phase board that receives the mains supply and feeds every BDB in the building, together with the water-pump motor and the EMDB. Its total connected load is:

**P = (BDB per unit × 8) + BDB of ground floor + Power of pump motor + EMDB + BDB of roof**

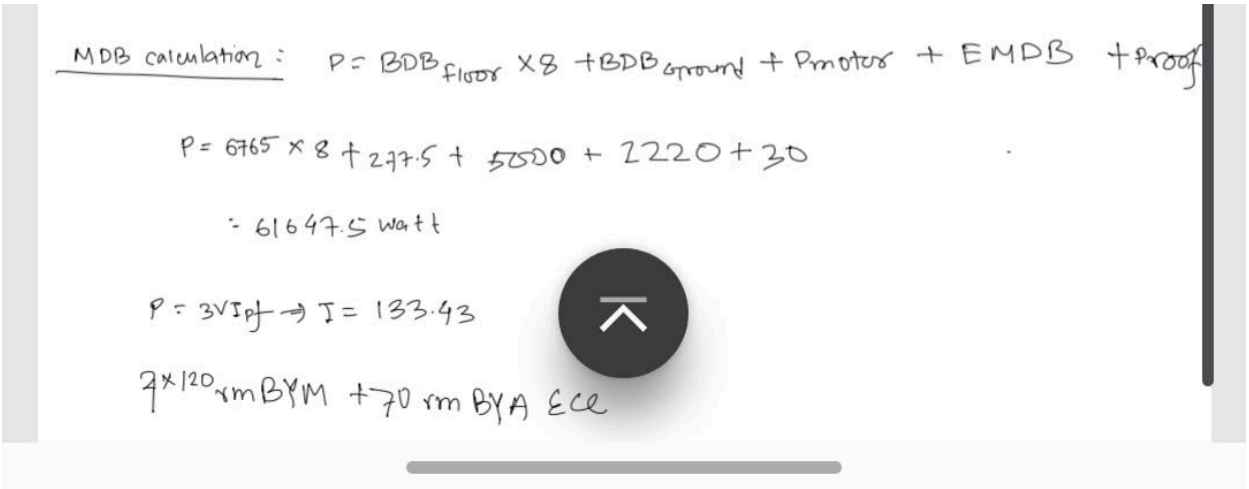
| Term | Value |
|---|---|
| BDB per unit × 8 units (6765 W × 8) | 54,120 W |
| BDB of ground (parking) floor | 277.5 W |
| Power of pump motor | 5000 W |
| EMDB (whole-building emergency load) | 2220 W |
| BDB of roof | 30 W |
| **Total connected load, P** | **61,647.5 W ≈ 61.6 kW** |

Since the supply is three-phase with the load split evenly across the R, Y and B phases, the phase current is:

**$I = P \div (3 \times V \times \cos\varphi) = 61{,}647.5 \div (3 \times 220 \times 0.7) = 61{,}647.5 \div 462 \approx 133.43$ A**

| Result | Value |
|---|---|
| Total connected load | 61.6 kW |
| Phase current | 133.43 A |
| Main incomer conductor | 4×120 mm² NYY + 70 mm² BYA (ECC) |
| Selected main breaker | 150 A MCCB |

The 150 A MCCB and 4×120 mm² NYY armoured incomer give a comfortable margin above the calculated 133.43 A phase current, allowing for the starting current of the pump motor and future load growth.




## 5. Distribution Boards and Single-Line Diagrams

The building's electrical system is arranged in a clear hierarchy of distribution boards. The Switch Boards (SB) group the electrical fixtures within each room and connect them through a limited number of final circuits. These circuits are then brought together at the Branch Distribution Boards (BDB), with separate

BDBs provided for each residential unit, the parking floor, and the rooftop. Finally, all of these BDBs are connected to the Main Distribution Board (MDB), which serves as the main point of distribution for the building.

A separate emergency system follows a similar arrangement, consisting of Emergency Switch Boards (ESB), Emergency Branch Distribution Boards (EBDB), and an Emergency Main Distribution Board (EMDB). This system is supplied by the standby generator and is used to maintain essential lighting throughout the building during a mains power failure. This section presents the single-line diagrams for all the distribution boards used in the building.

## 5.1 Switch Boards (SB)

Table 12 summarizes the three switch-board groups in the building, and Figures 7–9 show their single-line diagrams.

**Table 12: Switch board (SB) groups**

| Location | Circuits | Switch boards | Circuit MCB | Conductor |
|---|---|---|---|---|
| Typical unit | 4 | 8 | 5 A | 1.5 mm² |
| Garage / parking floor | 3 | 4 | 5 A | 1.5 mm² |
| Rooftop | 1 | 1 | 5 A | 1.5 mm² |

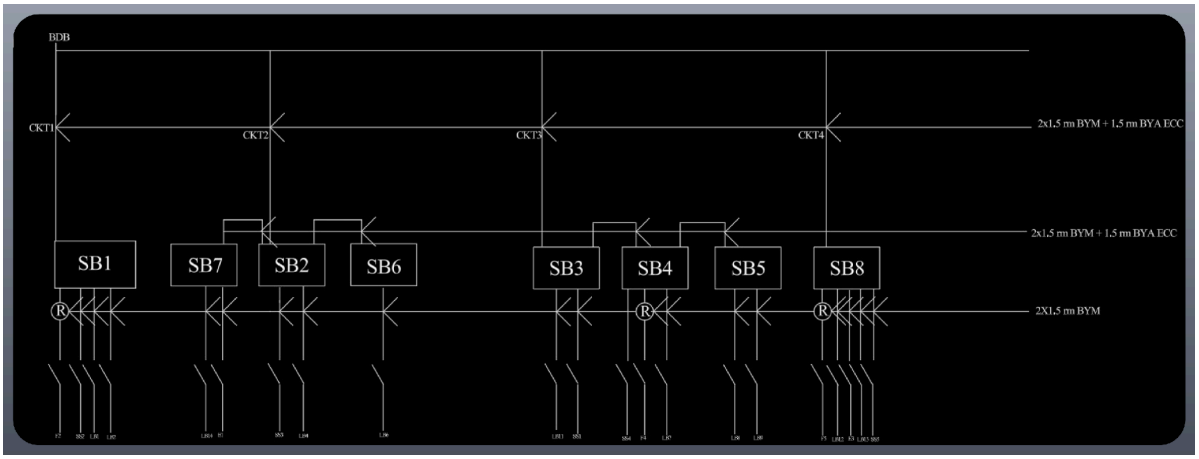



Figure 7: SB — single unit: eight switch boards (SB1–SB8) grouped onto four circuits.

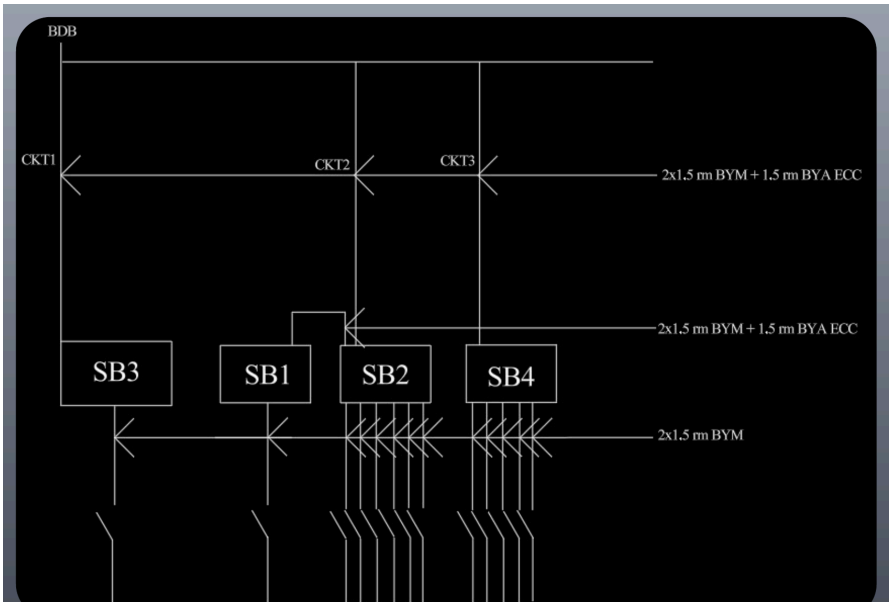



Figure 8: SB — garage: four switch boards grouped onto three circuits.

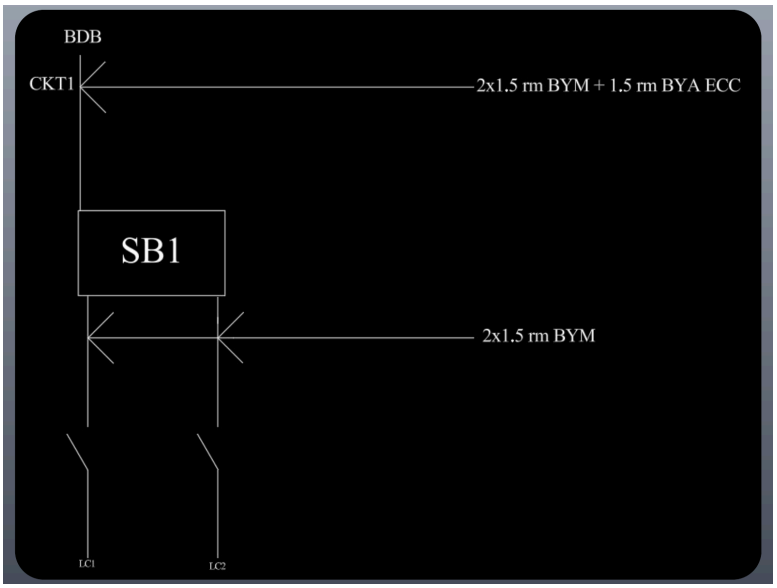



Figure 9: SB — rooftop: a single switch board on one circuit.

## 5.2 Branch Distribution Boards (BDB)

Each unit's BDB receives a 50 A-protected feeder from the MDB and splits it into four 5 A-protected lighting/fan circuits (CKT-1–CKT-4, feeding switch boards SB1–SB8) and one 50 A-protected socket-circuit group (P1–P4). The parking-floor and rooftop BDBs are simpler, each carrying only lighting/fan circuits. Table 13 summarizes all three, and Figures 10–12 show their single-line diagrams.

**Table 13: Branch Distribution Board (BDB) summary**

| Board | Incomer (from MDB) | Final circuits | Final conductor |
|---|---|---|---|
| BDB — typical unit | 50 A MCCB, 2×25 mm² BYM + 25 mm² BYA ECC | 4× lighting (5 A SP MCCB) + 1× socket group (50 A SP MCCB) | 2×1.5 mm² BYM +1.5 mm² BYA ECC (lighting); 2×6 mm² BYM + 6 mm² BYA ECC (sockets) |
| BDB — parking floor | 5 A MCCB, 2×1.5 mm² BYM + 1.5 mm² BYA ECC | 3× lighting (5 A SP MCCB) | 2×1.5 mm² BYM + 1.5 mm² BYA ECC |
| BDB — rooftop | 5 A MCCB (single-phase, SP), 2×1.5 mm² BYM +1.5 mm² BYA ECC | 1× lighting circuit | 2×1.5 mm² BYM |

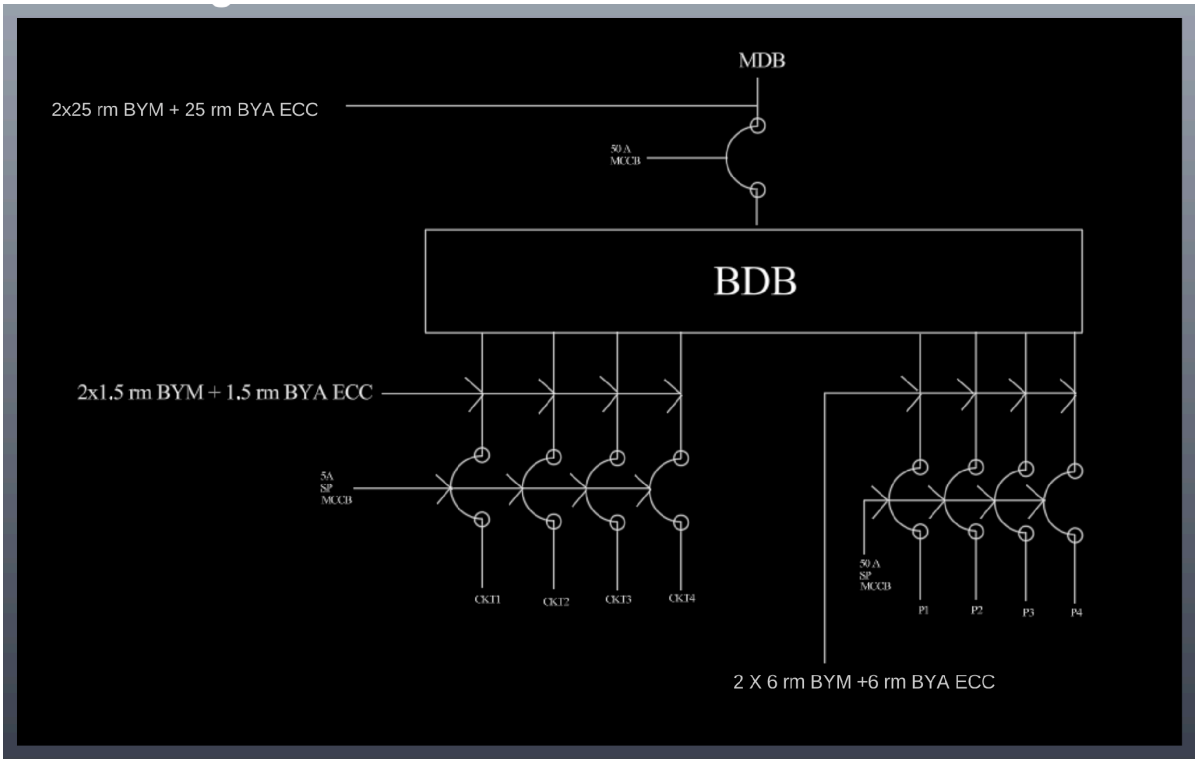



Figure 10: BDB — single unit: incomer from the MDB, four lighting circuits (CKT1–CKT4) and one socket-circuit group (P1–P4).

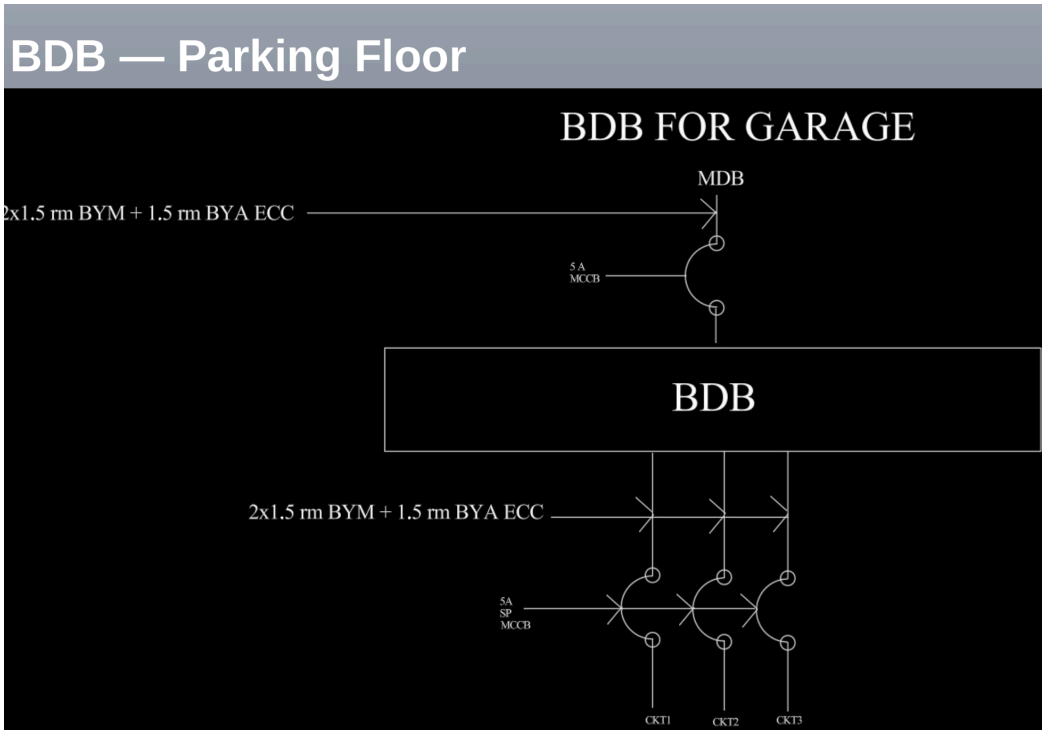



Figure 11: BDB — parking floor: three lighting circuits (CKT1–CKT3).

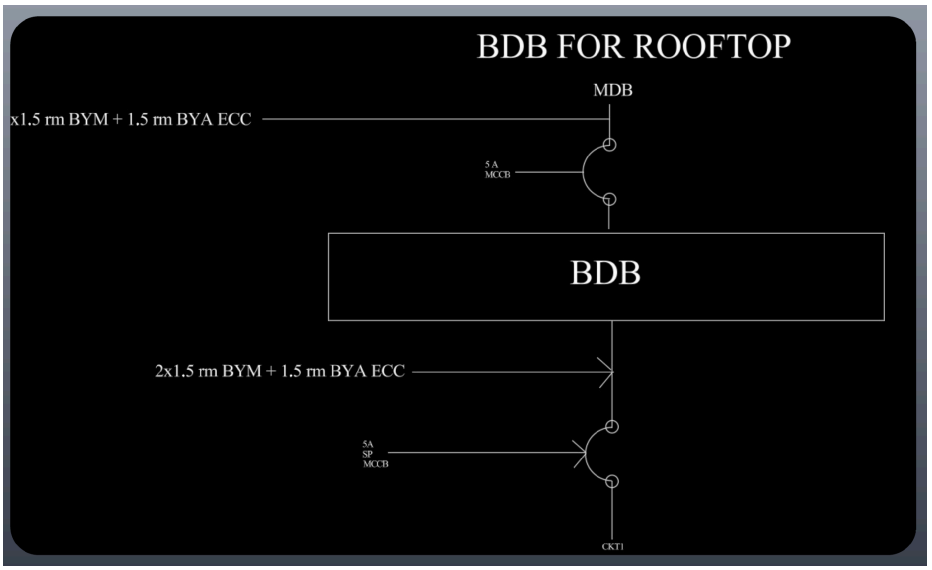



Figure 12: BDB — rooftop: a single lighting circuit.

### 5.3 Emergency Switch Boards (ESB)

The emergency switch boards mirror the normal switch boards but carry only the lighting points and exhaust/ceiling fans designated as essential loads. Figures 13–15 show the ESB single-line diagrams for a typical unit, the garage, and the lobby.

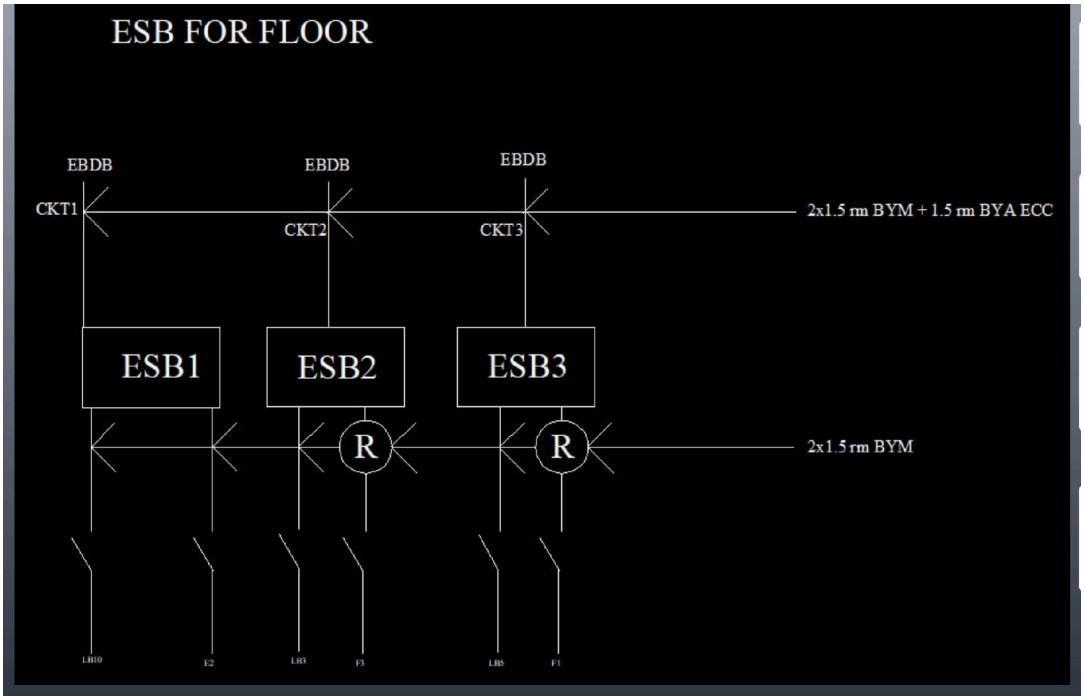



Figure 13: ESB — single unit: three emergency switch boards (ESB1–ESB3) on three circuits.

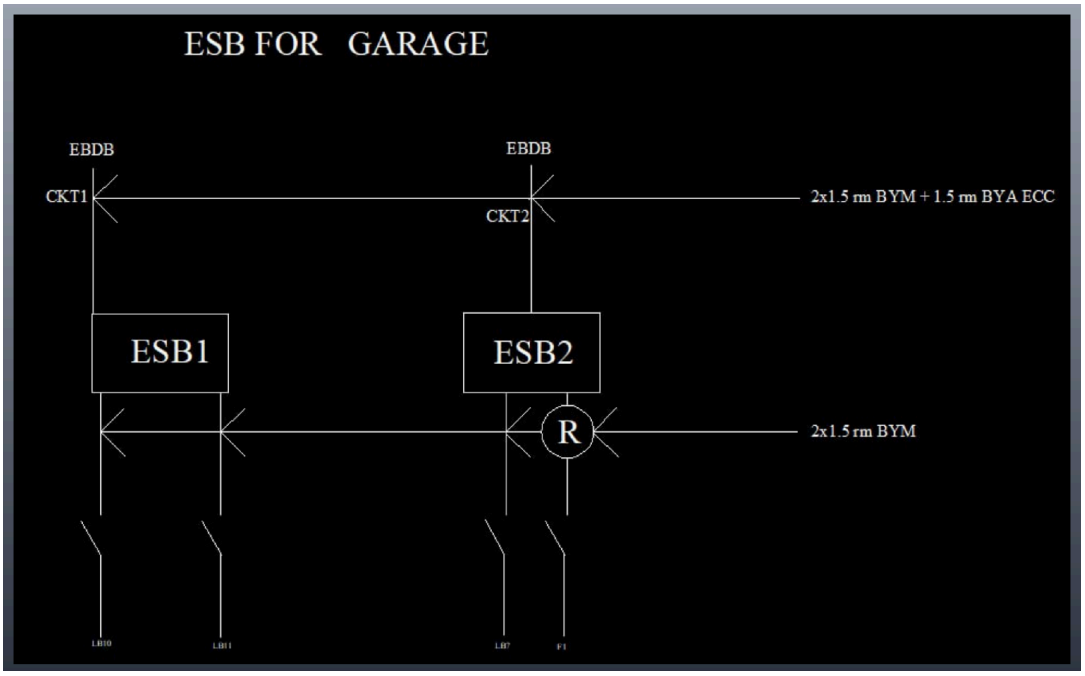



Figure 14: ESB — garage: two emergency switch boards on two circuits.

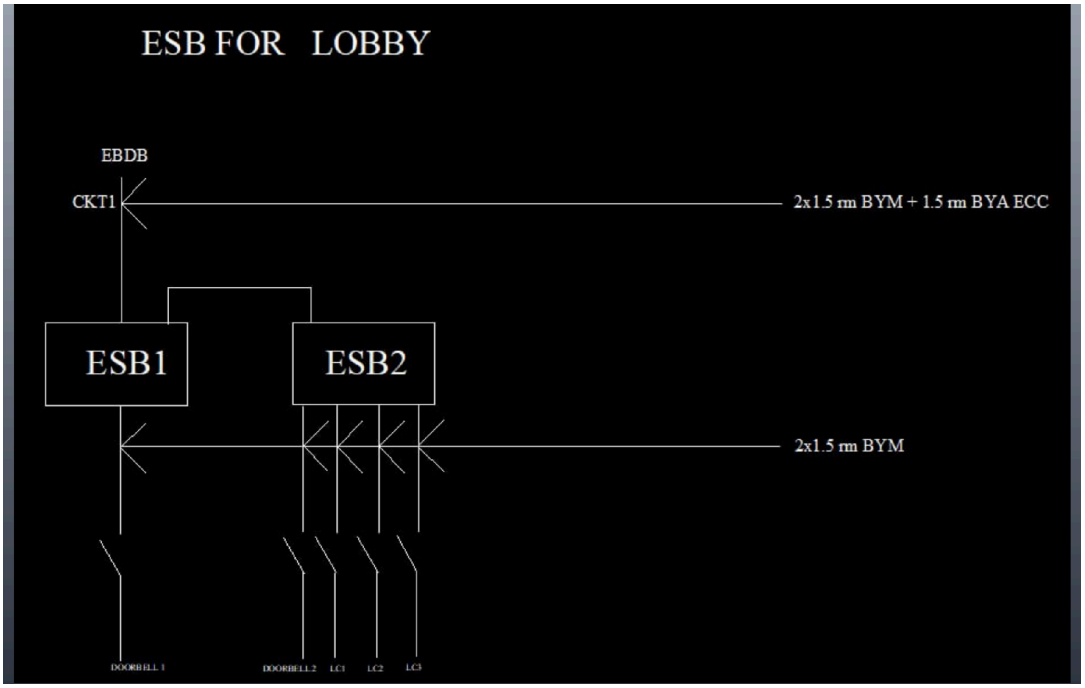



Figure 15: ESB — lobby: two emergency switch boards (doorbells and lobby light points) on one circuit.

## 5.4 Emergency Branch Distribution Board (EBDB)

Every unit, the garage and the lobby each have their own EBDB, fed from the EMDB through a 40 A MCCB and wired in 2×1.5 mm² BYM + 1.5 mm² BYA (ECC). As calculated in Section 4.4, the per-unit emergency design load is 232.5 W at a demand factor of 0.75, drawing about 1.50 A. Figure 16 shows the EBDB arrangement for a typical floor and for the garage.

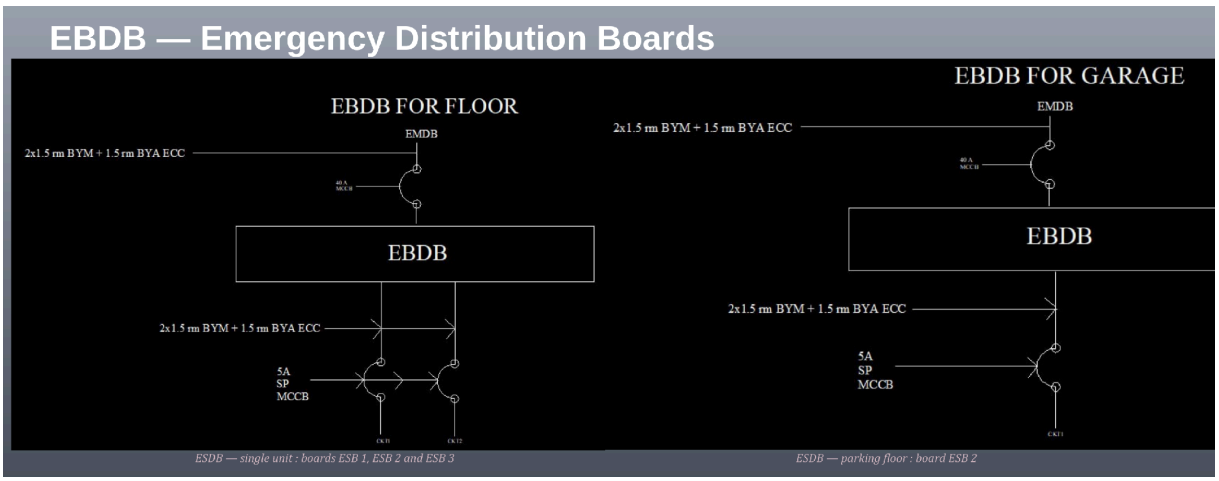



Figure 16: EBDB — emergency distribution boards for a typical floor (ESB1–ESB3) and for the garage (ESB2).

## 5.5 Main Distribution Board (MDB)

The MDB is the building's three-phase incomer board. It receives the utility supply through a 150 A MCCB and distributes it, phase-balanced across R, Y and B, to the eight unit BDBs, the ground-floor and

lobby BDBs, the water-pump motor, and — via the LT meter and an automatic changeover arrangement with the standby generator — to the EMDB. Figure 17 shows the complete building single-line diagram.

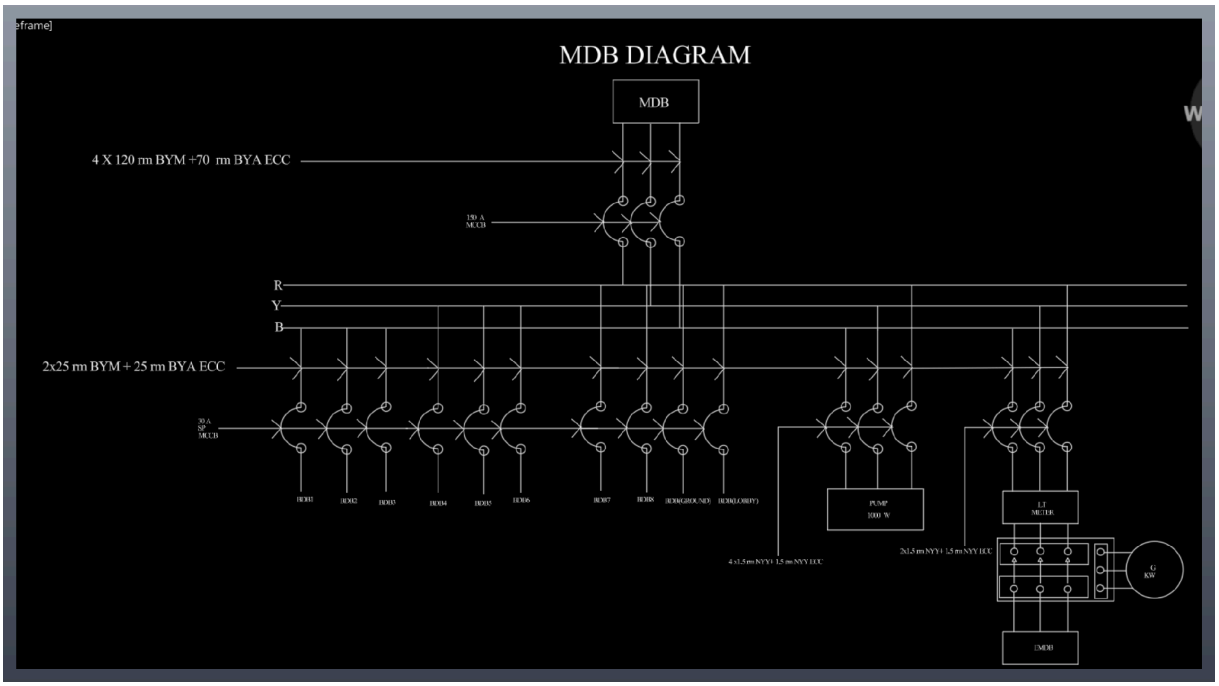



Figure 17: MDB — building single-line diagram: three-phase incomer feeding all branch boards, the pump and the standby-generator changeover to the EMDB.

## 5.6 Emergency Main Distribution Board (EMDB)

The EMDB is energised from the standby generator whenever the normal mains supply fails, and feeds every EBDB in the building — one per unit, plus the ground floor and the lobby — through individually protected 5 A circuits. Figure 18 shows the EMDB single-line diagram.

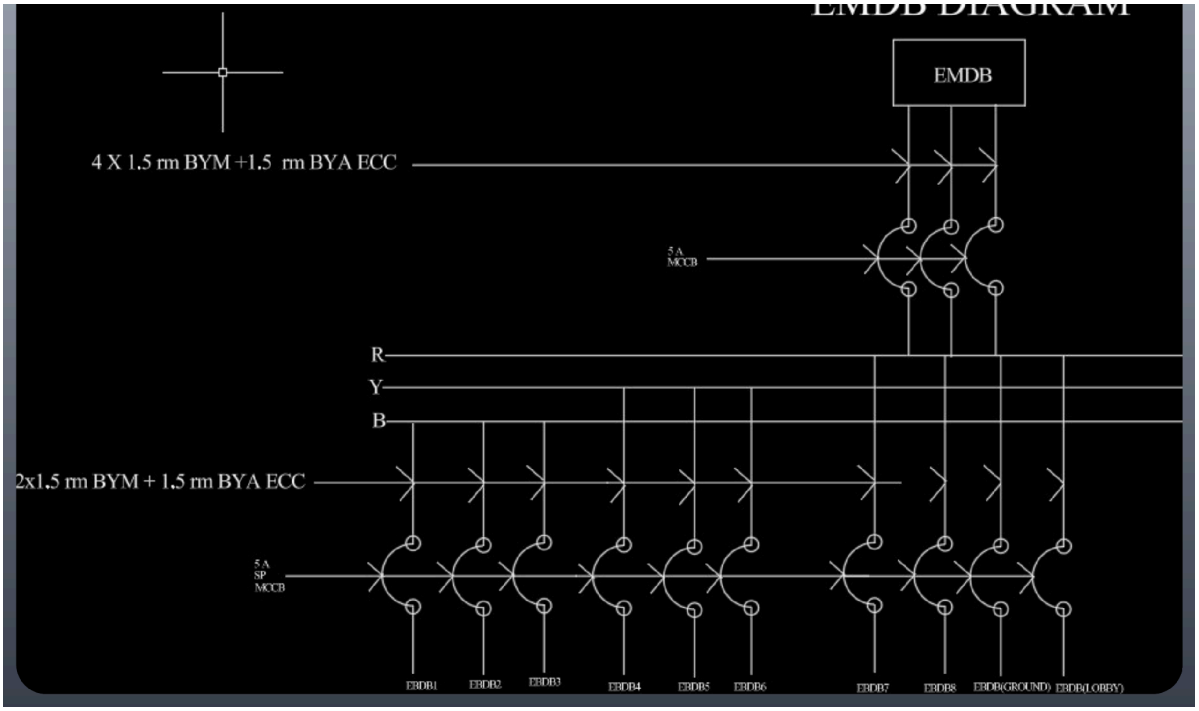


Figure 18: EMDB — building single-line diagram: three-phase incomer from the standby generator feeding all emergency branch boards.

# 6. Conduit and Cable Selection

Two cable types are used throughout the building. BYM (PVC-insulated, non-sheathed single-core conductor) carries the phase/line conductors of every internal circuit, and the same non-sheathed conductor type, BYA — rated 450/750 V to BS 6004:1975 — is used for every earth-continuity conductor (ECC), so that ECC size can be quoted independently of the phase conductor. NYY (PVC-insulated and PVC-sheathed, rated 600/1000 V) is reserved for the single heavy-duty run that matters most for mechanical protection: the three-phase main incomer from the utility supply to the MDB. Every conductor size below was cross-checked against standard single-core PVC cable current-rating and voltage-drop tables (as tabulated in the Manual of Eastern Cables / BICC Cables and Tables for international standard conductor sizes) to confirm it carries its circuit's design current with an adequate margin.

## 6.1 Point-Wiring Conduits

Within a room, several fixture points sharing one switch board are often run together in a single conduit rather than one conduit per point. Three standard point-wiring conduits are used, sized by how many 1.5 mm² BYM conductors they carry, as listed in Table 14.

**Table 14: Point-wiring conduit specification**

| Conduit | Specification | Typical use |
|---|---|---|
| C1 | 2 × 1.5 mm² BYM | A single light or fan point wired back to its switch board |
| C2 | 4 × 1.5 mm² BYM | Two point-wiring runs sharing one conduit |
| C3 | 6 × 1.5 mm² BYM | Three point-wiring runs sharing one conduit |
| C10 | 2 × 6 mm² BYM + 6 mm² BYA (ECC) | One 3-pin, 15 A socket-outlet feeder |
| C11 | 4 × 6 mm² BYM + 6 mm² BYA (ECC) | Two 3-pin, 15 A socket-outlet feeders sharing one conduit |

## 6.2 Circuit and Feeder Conductors

Table 15 summarizes the conductor and protective-device size selected at every level of the distribution hierarchy, drawing together the calculations of Section 4 and the single-line diagrams of Section 5.

**Table 15: Conductor selection summary — normal and emergency systems**

| Feeder / circuit | Protection | Conductor |
|---|---|---|
| Unit lighting/fan circuit (CKT-1–4) | 5 A SP MCB | 2×1.5 mm² BYM + 1.5 mm² BYA ECC |
| Unit socket-circuit group (P1–P4) | 50 A SP MCCB | 2×6 mm² BYM + 6 mm² BYA ECC |
| Unit BDB incomer (from MDB) | 50 A MCCB | 2×25 mm² BYM + 25 mm² BYA ECC |
| Parking-floor / rooftop BDB incomer | 5 A MCCB | 2×1.5 mm² BYM + 1.5 mm² BYA ECC |
| Unit EBDB incomer (from EMDB) | 40 A MCCB | 2×1.5 mm² BYM + 1.5 mm² BYA ECC |

| Feeder / circuit | Protection | Conductor |
|---|---|---|
| EMDB main incomer | 5 A MCCB | 4×1.5 mm² BYM + 1.5 mm² BYA ECC |
| MDB main incomer (from utility) | 150 A MCCB | 4×120 mm² NYY + 70 mm² BYA ECC |

The two largest feeders — the unit BDB incomer (43.92 A design current, 2×25 mm² BYM) and the MDB main incomer (133.43 A design current, 4×120 mm² NYY) — were sized directly from the load calculations in Sections 4.3 and 4.5, then rounded up to the next standard conductor size with an adequate current-carrying margin. All smaller lighting and emergency circuits use the minimum practical size, 1.5 mm², since their currents (well under 3 A per circuit) are far below the rating of even the smallest standard conductor; the deciding constraint for these is mechanical robustness rather than current-carrying capacity.

# 7. Auxiliary Systems

Two auxiliary systems were designed alongside the power wiring: a telephone/television/data network for each unit, and a lightning protection system for the building as a whole.

## 7.1 Telephone, Television and Data Network

Each unit is provided with a telephone point (TP1), a television point (TV1) and a data/network point (DT1), each fed by the cable type appropriate to its signal, as listed in Table 16. Figure 19 shows the conduit routing for these points within a single unit, and Figure 20 shows how each point is fed back to its respective incoming service.

**Table 16: Telephone, TV and data point cabling**

| Point | Service | Cable |
|---|---|---|
| TP1 | Telephone | PVC-insulated and sheathed twisted-pair telecom cable |
| TV1 | Television (dish) | 75 Ω impedance coaxial cable |
| DT1 | Data / network | CAT5 / CAT6 cable |

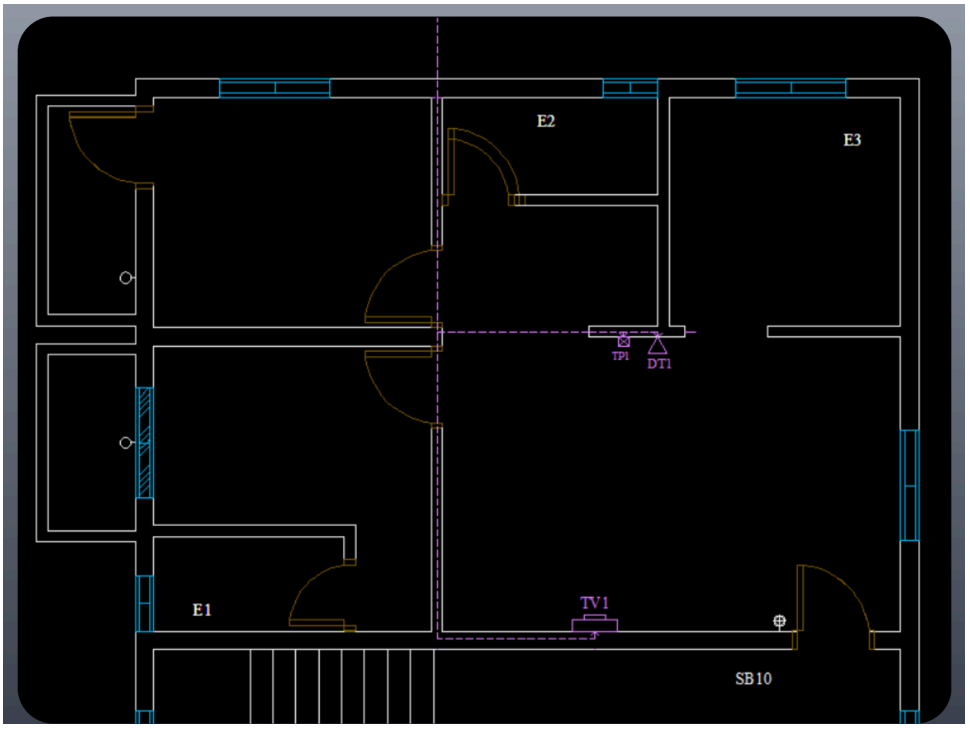



Figure 19: Conduit routing for the telephone (TP1), television (TV1) and data (DT1) points — single unit.

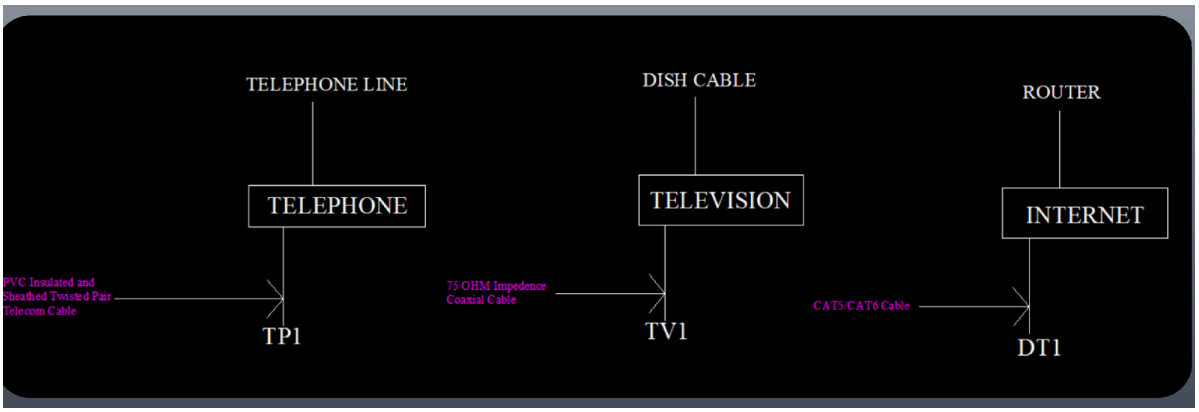



Figure 20: Distribution arrangement for the telephone line, dish TV cable and router/internet feed.

## 7.2 Lightning Protection System

The roof is fitted with a lightning protection system made up of air conductors (roof-level rods that intercept a strike), down conductors (which carry the strike current to earth) and roof conductors (which link the air conductors together across the roof), following the symbols in Figure 21.

An air conductor is placed at each of the building's four roof corners, and the maximum allowable spacing between adjacent air conductors is taken as 25 ft. Checking the roof dimensions against this spacing rule:

| Check | Calculation | Result |
|---|---|---|
| Building perimeter | 2 × (61′ + 35.625′) | 193.25 ft |
| Additional air conductors along the width (35.625′) | ⌈35.625 / 25⌉ − 1 | 1 |
| Additional air conductors along the length (61′) | ⌈61 / 25⌉ − 1 | 2 |
| Minimum down conductors by perimeter rule | ⌈193.25 / 100⌉ | 2 |
| Down conductors actually provided | increased for symmetry | 4 |

Although the perimeter rule requires only two down conductors, four were provided — one at each corner of the building, in line with the four corner air conductors — so that the down-conductor network is symmetric and every corner has its own independent path to earth.

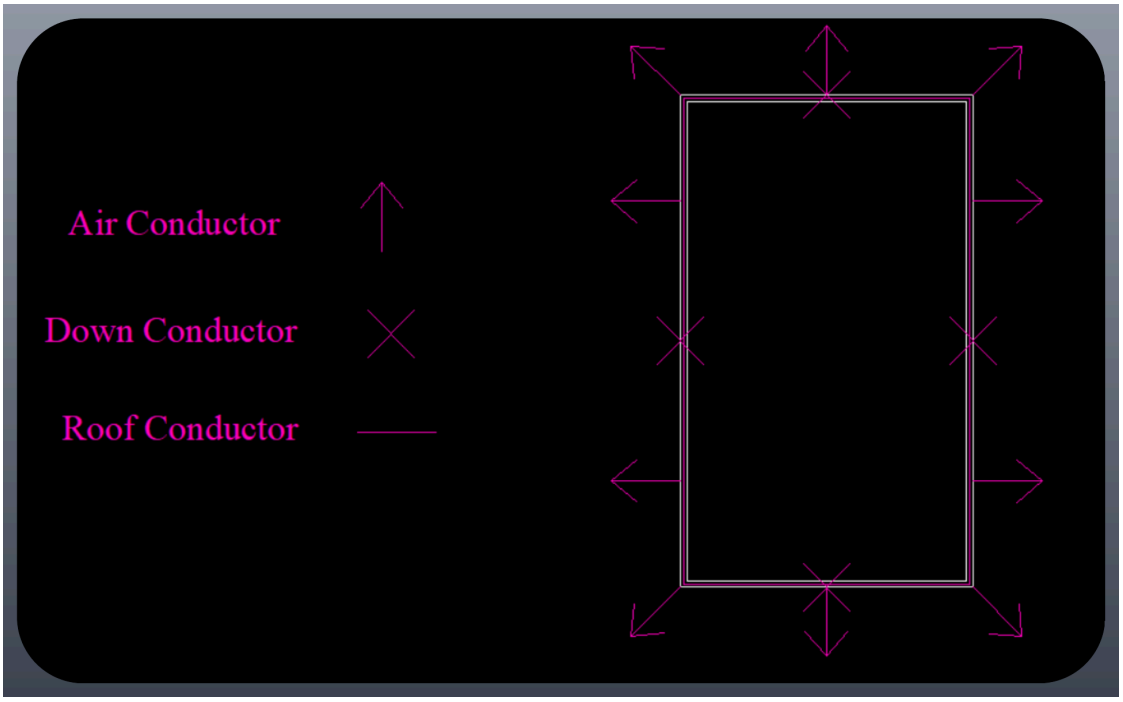



Figure 21: Lightning protection layout — air, down and roof conductor placement.

## 8. Summary Table

Table 17 draws together the building-wide totals developed in the preceding sections into a single overview of the completed design.

**Table 17: Building-wide design summary**

| Item | Total |
|---|---|
| Residential units | 8 (2 per floor × 4 floors) |
| Light points (LB / CB / TB) | 112 |
| Ceiling fans | 40 |
| Exhaust fans | 24 |
| Switch boards (SB) | 80 |

| Item | Total |
|---|---|
| Switchboard sockets (SS, 2-pin) | 40 |
| 3-pin, 15 A socket outlets (P) | 32 |
| Total socket outlets (SS + P) | 72 |
| Branch / distribution boards, normal + emergency (SB, BDB, ESB, EBDB, MDB, EMDB) | 21 |
| Total connected load | 61.6 kW (61,647.5 W) |
| MDB phase current | 133.43 A |
| Main breaker | 150 A MCCB |
| Main incomer cable | 4×120 mm² NYY + 70 mm² BYA (ECC) |
| Typical unit BDB incomer cable | 2×25 mm² BYM + 25 mm² BYA (ECC) |
| Typical lighting/fan circuit cable | 2×1.5 mm² BYM + 1.5 mm² BYA (ECC) |
| Typical socket-circuit cable | 2×6 mm² BYM + 6 mm² BYA (ECC) |
| System voltage | 220 V, single-phase per circuit; 3-phase, 4-wire (R-Y-B-N) at the MDB |
| Emergency system design load (EMDB) | 2220 W, generator-backed |
| Lightning-protection down conductors | 4 (corner-mounted) |

## 9. Conclusion

This report presented the complete electrical layout design of a four-storied residential building for the EEE 4418 course, covering everything from the initial architectural planning to the final selection of cables and conduits. The entire design was prepared in AutoCAD. The building consists of eight residential units arranged as mirrored pairs across four typical floors, with a parking and service area on the ground floor. Since all the residential units follow the same basic design, the electrical arrangement remains consistent throughout the building. Each unit requires 14 light points, 5 ceiling fans, 8 switch boards, and 4 socket outlets, resulting in a total of 112 light points, 40 fans, 80 switch boards, and 72 socket outlets for the whole building.

The electrical fittings and fixtures were arranged based on practical design considerations. Each room was provided with a switch board, the kitchens were positioned along the outer walls to allow proper ventilation, and a dedicated 3-pin socket was included in every room for heavier appliances. All the fixtures were also included in the connected-load calculations. Starting from the individual final circuits, where the current ranged from 1.4 A to 2.5 A, the loads were gradually combined at each unit's Branch Distribution Board, giving a maximum unit current of 43.92 A, and finally brought together at the Main Distribution Board. Based on these calculations, the building has a total connected load of 61.6 kW, with a corresponding phase current of 133.43 A. A 150 A main breaker was therefore selected for the incoming supply.

The distribution system also includes a separate generator-backed emergency network to ensure that essential services remain available during a mains power failure. This network consists of Emergency Switch Boards (ESB), Emergency Branch Distribution Boards (EBDB), and an Emergency Main

Distribution Board (EMDB). Through this arrangement, essential lighting and exhaust points can continue to operate even when the normal power supply is unavailable. Cable and conduit sizes were selected according to the calculated design currents, ranging from 1.5 $mm^2$ conductors for smaller lighting circuits up to a 120 $mm^2$ conductor for the main incomer. Standard cable current-rating tables were used as a reference to ensure that each selected conductor could safely handle its required load.

Finally, the overall design was completed with two auxiliary systems: a telephone/television/data network and a lightning protection system. The communication network was planned to serve all eight residential units, while the lightning protection system was designed according to the roof perimeter, keeping the spacing of the air conductors within the 25 ft limit. Four down conductors were provided symmetrically at the corners of the building to create independent paths to earth. Taken together, the electrical distribution, emergency supply, communication network, and lightning protection system form a complete and consistent design that can be developed further into construction drawings and used for practical site implementation.